



TRAINING PROGRAMME

Azure DevOps Engineer

Plan, build, release, secure and monitor software on Azure with Azure DevOps and GitHub

45 HOURS

FOR Azure administrators and developers who build delivery pipelines on Azure

PREPARES FOR Microsoft Certified: DevOps Engineer Expert (AZ-400)



Mohamed Aziz BEN HAH

mohamedazizbenhaha.netlify.app

Overview

This programme teaches participants to run DevOps practices on Microsoft Azure and prepares the AZ-400 exam for the DevOps Engineer Expert certification. They plan with Azure Boards and GitHub Projects, manage code in Azure Repos and GitHub, build tested YAML pipelines with container builds and package feeds, provision infrastructure with Bicep and Terraform, release to AKS progressively with approvals, secure pipelines without stored credentials, and close the loop with Application Insights. Exam practice closes the content, and the programme ends with a complete Azure DevOps project built and presented by each team.

At a glance

#	PHASE	FOCUS	HOURS
01	Planning & Source Control	Azure Boards, GitHub Projects, flow metrics, Azure Repos and GitHub, branch policies	6 h
02	Continuous Integration	YAML pipelines, templates, container builds to ACR, Azure Artifacts, GitHub Actions	9 h
03	Infrastructure & Deployment	Bicep and Terraform, configuration management, AKS, release strategies	9 h
04	Security, Monitoring & Exam Preparation	Key Vault, workload identity federation, Advanced Security, Application Insights, AZ-400	9 h
05	Final Project	A complete Azure DevOps project, presented to the group	6 h
	Buffer	Catch-up, revision, extra lab time	6 h
	Total		45 h

Key facts

AUDIENCE	Azure administrators and developers who build delivery pipelines on Azure
PREREQUISITES	Azure administration (course: Azure Administrator) or CI/CD basics (course: DevOps Engineering); not taught again. The Expert certification also requires Azure Administrator or Azure Developer Associate
DURATION	45 h: 39 h of chapters + 6 h buffer
FORMAT	Short lectures followed by guided labs; one application carried through every phase; exam practice
LAB ENVIRONMENT	One laptop per participant (Git, Azure CLI, VS Code, Docker); per team, an Azure DevOps and a GitHub organisation (Advanced Security enabled) and an Azure subscription with a budget alert, provided by the organiser; resources deleted after each lab
ASSESSMENT	A lab validated at the end of each phase, practice exam questions and a final project presented to the group; certificate of completion for participants who complete the labs and the project
OUTCOME	Participants can take an application from a planned backlog item to a monitored canary release on AKS through secure pipelines, and sit the AZ-400 exam

PHASE 01

6 H

Planning & Source Control

Participants plan the work for the course application with Azure Boards and GitHub Projects and measure its flow, then manage the code in Azure Repos and GitHub with a branching strategy, pull requests and branch policies. Validated lab: a backlog item linked to a branch, a reviewed pull request and a protected main branch.

#	CHAPTER	CONTENT	FORMAT	HOURS
1.1	Planning	Azure Boards, Agile planning, GitHub Projects, flow metrics, dashboards; wikis, Markdown and Mermaid documentation, release notes, webhooks	Lecture + Lab	3 h
1.2	Source Control	Azure Repos and GitHub, branch strategies, pull requests and branch policies, repository permissions, Git LFS, recovering and purging data	Lecture + Lab	3 h

PHASE 02

9 H

Continuous Integration

Participants build the course application with Azure Pipelines in YAML, factor the pipeline into templates, build and push container images to Azure Container Registry, publish versioned packages to Azure Artifacts, and run the same build from GitHub Actions to Azure. Validated lab: a pull request validated by a templated pipeline, then the merge publishing a versioned image to ACR and a package to a feed.

#	CHAPTER	CONTENT	FORMAT	HOURS
2.1	Azure Pipelines	YAML pipelines, agents and pools, triggers, stages, jobs and steps; testing strategy, code coverage, quality gates	Lecture + Lab	3 h
2.2	Templates & Container Builds	Pipeline templates, Azure Container Registry, multi-stage builds in pipelines	Lecture + Lab	3 h
2.3	Package Management	Azure Artifacts and GitHub Packages, feeds, views and upstream sources, SemVer versioning; GitHub Actions to Azure with OpenID Connect; pipeline health, retention, classic-to-YAML migration	Lecture + Lab	3 h

PHASE 03

9 H

Infrastructure & Deployment

Participants provision the course environment with Bicep and Terraform and keep machine configuration in a desired state, create an AKS cluster and deploy to it from pipelines, then release safely with environments, approvals and checks, blue-green and canary deployments and feature flags. Validated lab: a canary release on AKS promoted after an approval, with the infrastructure created by the pipeline.

#	CHAPTER	CONTENT	FORMAT	HOURS
3.1	Infrastructure as Code	Bicep, Terraform on Azure, desired state configuration with Azure Machine Configuration (Automation State Configuration as the legacy service); Azure Deployment Environments	Lecture + Lab	3 h
3.2	AKS	Provisioning, deploying from pipelines, scaling	Lecture + Lab	3 h
3.3	Release Strategies	Environments, approvals and checks, blue-green, canary, deployment slots, feature flags with Azure App Configuration	Lecture + Lab	3 h

PHASE 04

9 H

Security, Monitoring & Exam Preparation

Participants remove secrets from pipelines with Key Vault and service connections that use workload identity federation, scan code and dependencies with GitHub Advanced Security for Azure DevOps and Defender for Cloud, then close the feedback loop with Application Insights, KQL, dashboards and alerts before practising AZ-400 questions. Validated lab: a pipeline that deploys with no stored credentials, a push containing a secret blocked by push protection, and an alert on the application's failure rate.

#	CHAPTER	CONTENT	FORMAT	HOURS
4.1	Secure Pipelines	Key Vault, service connections with workload identity federation, permissions, GitHub Advanced Security for Azure DevOps, Defender for Cloud DevOps security, GitHub Apps and GITHUB_TOKEN, Azure DevOps security groups	Lecture + Lab	3 h
4.2	Monitoring & Feedback	Application Insights, Container insights, KQL, dashboards, alerts	Lecture + Lab	3 h
4.3	AZ-400 Preparation	Practice questions and review	Lecture + Workshop	3 h

PHASE 05

6 H

Final Project

Teams of two or three deliver a provided application from backlog to production on Azure, reusing the phase labs, then present the result as they would to an engineering manager.

#	CHAPTER	CONTENT	FORMAT	HOURS
5.1	Brief & Build	Read the brief, set up the Boards backlog, Repos with branch policies, a YAML pipeline to ACR and the Bicep or Terraform infrastructure	Project	3 h
5.2	Release & Monitor	AKS deployment with approvals and canary, secrets from Key Vault, Application Insights; README and architecture diagram	Project	1.5 h
5.3	Demo & Review	Live demo from backlog item to canary release, cleanup; questions and feedback	Project	1.5 h

FINAL PROJECT

From Backlog to Canary Release on Azure

A backlog item planned in Azure Boards, coded under branch policies, built by a YAML pipeline to ACR, deployed to AKS on infrastructure created with Bicep or Terraform, released as a canary after approval, with secrets in Key Vault and health watched in Application Insights.

STACK Azure Boards, Azure Repos or GitHub, Azure Pipelines, Azure Container Registry, Bicep or Terraform, AKS, Key Vault, Application Insights

DELIVERABLE Repositories with pipeline and infrastructure code, architecture diagram, live demo